FINITE-HORIZON BLOCK AND SLIDING-WINDOW RUIN FOR BROWNIAN RISK MODELS

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Abstract. We introduce and investigate sliding-window ruin and block ruin for general risk processes over a fixed horizon. Our analysis focuses on the classical benchmark Brownian risk model and yields exact asymptotics of the corresponding ruin probabilities. We also consider perturbations by a compound-Poisson process with non-negative subexponential claims, Brownian motion with exponential resetting and simultaneous block ruin for correlated bivariate Brownian risks.

1. Introduction

Let $X(t), t \geq 0$ be a stochastic process with càdlàg paths modelling the claim surplus of a given insurance portfolio, with $X(0) = 0$. Writing $R_u(t) = u - X(t)$ for the reserve process with initial capital $u > 0$, the classical finite-time ruin probability

$$\psi(u,T) := \mathbb{P}\left\{ \sup_{0\leq t\leq T} X(t) > u \right\}$$

quantifies the probability that the reserve becomes negative at least once by time $T > 0$.
In order to distinguish an isolated crossing from distress recurring across monitoring periods, we introduce two new ruin functionals related to exceedances over prescribed reporting blocks or sliding windows.
Throughout we fix $h > 0$ and an integer $m \geq 2$ and put $T = mh$; the *block ruin probability* is

$$p_h^{\mathrm{bl}}(u,T) := \mathbb{P}\left\{ \bigcap_{k=1}^{m} \left\{ \sup_{t\in[(k-1)h,kh]} X(t) > u \right\} \right\} \tag{1}$$

and thus ruin is required to occur at least once in each of the m reporting blocks. The closely related *sliding-window ruin probability* is then defined by

$$p_h^{\mathrm{sl}}(u,T) := \mathbb{P}\left\{ \inf_{t\in[0,T-h]} \sup_{s\in[t,t+h]} X(s) > u \right\}. \tag{2}$$

For X with continuous paths the moving-window maximum is continuous in the window start and hence (2) means that every length-h window contains a ruin epoch, or equivalently that there is no recovery interval of length h on which $X \leq u$. For discontinuous paths we retain the strict infimum in (2): it requires a positive margin uniform over all window starts and need not be equivalent to a strict exceedance in each window separately. The reporting blocks are closed, so a crossing at a shared endpoint can satisfy both adjacent block requirements. For càdlàg paths, the moving-window supremum is right-continuous in its start, so the infimum in (2) can be taken over $(\mathbb{Q} \cap [0, T-h)) \cup \{T-h\}$. Each fixed-window supremum is measurable,

Date: November 2, 2026.
2020 *Mathematics Subject Classification.* Primary 60G70; Secondary 60J65, 60G15, 91G05.
Key words and phrases. Brownian risk model; block ruin; sliding-window ruin; simultaneous ruin; first-passage time; stochastic resetting; compound-Poisson process; subexponential distribution; exact asymptotics.

since it is the supremum over rational times before the right endpoint together with the value at that endpoint. Thus (2) defines a measurable event.

Our asymptotic analysis focuses on the classical benchmark Brownian risk model

$$X(t) = B(t) - ct, \qquad t \geq 0,$$

where $B(t), t \geq 0$ is standard Brownian motion and $c > 0$ is the premium rate. The reserve process is then $R_u(t) = u + ct - B(t), t \geq 0$.

The sliding requirement is related to classical Parisian ruin but has a different meaning. Classical Parisian ruin requires the reserve to remain below zero throughout an interval of prescribed length. Parisian ruin for spectrally negative Lévy risk processes is studied in [3, 20]; Gaussian counterparts for self-similar models and over finite horizons are treated in [4, 5]. Parisian ruin for Brownian risk models over an infinite horizon is studied in [1]. Discrete monitoring of Brownian ruin and its Parisian variants is considered in [18]. Further Gaussian models include random premium income [22], power-asymmetric variance near the optimal point [16], and locally self-similar processes [21]. Our sliding event instead excludes sustained recovery; its post-hit constant has the Parisian-type survival representation in (11).

Our main results yield the following asymptotic equivalents as $u \to \infty$:

$$p_h^{\mathrm{bl}}(u, mh) \sim 2G_m(h, c)\Psi\left(\frac{u + ch}{\sqrt{h}}\right), \qquad p_h^{\mathrm{sl}}(u, mh) \sim 2G_m^{\mathrm{sl}}(h, c)\Psi\left(\frac{u + ch}{\sqrt{h}}\right),$$

with Ψ denoting the standard normal upper tail.

Conditional on a first-block exceedance, the first hit of u approaches h. The strong Markov property at this hit reduces the remaining requirements to finite-level Brownian events and yields the constants G_m and G_m^{sl}.

Related simultaneous-crossing problems belong to the theory of Gaussian conjunctions. Exact asymptotics for finite-horizon conjunctions of independent Gaussian coordinates are obtained in [7]. For correlated Brownian risks, infinite-horizon orthant hitting is studied in [6] and finite-horizon bivariate simultaneous ruin in [8]. Finite-horizon simultaneous ruin for classes of multivariate Gaussian risk models with stationary increments is treated in [2]. Related simultaneous-ruin asymptotics for independent gamma-reflected Brownian motions are obtained in [17]. These events require the coordinates to cross their respective levels at a common time; our block event requires such a simultaneous crossing in every reporting block.

We then consider three extensions. For a compound-Poisson risk model perturbed by Brownian motion and with non-negative subexponential claims, both repeated-ruin probabilities are asymptotic to $\lambda h \bar{F}(u)$ because one sufficiently large first-block claim dominates. We next consider the Brownian resetting model of [9] with both initial value and reset level at zero. The entire drifted process restarts at independent exponential times. The first-block Gaussian cost remains and the later-block factor accounts for the reset clock up to the last required crossing. For bivariate Brownian risks with correlation ρ and thresholds u, au, we treat simultaneous block ruin in the binding regime $\rho < a \leq 1$. The first-block contribution is the bivariate finite-time ruin asymptotic; later blocks yield a finite-level simultaneous-crossing factor.

The paper is organised as follows. In Section 2 we present the main results for the Brownian risk model, followed by the extensions and discussion in Section 3. Numerical illustrations of the Brownian and resetting results are given in Section 4; proofs are relegated to Section 5.

2. Main results

In this section we take $X(t) = B(t) - ct$ with $c > 0$, fix h, m, T as specified in the Introduction and let

$$\Psi(x) := \mathbb{P}\{Z > x\}, \qquad \varphi(x) := \frac{1}{\sqrt{2\pi}} \exp(-x^2/2), \qquad x \in \mathbb{R},$$

where Z is a standard normal random variable.
The classical finite-time ruin probability is explicitly known: for $u \geq 0$,

$$\psi(u, T) = \mathbb{P}\left\{ \sup_{0 \leq t \leq T} X(t) > u \right\} = \Psi\left(\frac{u + cT}{\sqrt{T}} \right) + \exp(-2cu) \Psi\left(\frac{u - cT}{\sqrt{T}} \right). \tag{3}$$

The following elementary comparison explains the ordering of the ruin functionals.

Lemma 2.1. *For every $u > 0$*

$$\mathbb{P}\left\{ \bigcap_{k=1}^{m-1} \{X(kh) > u\} \right\} \leq p_h^{\mathrm{sl}}(u, T) \leq p_h^{\mathrm{bl}}(u, T) \leq \mathbb{P}\left\{ \sup_{0 \leq t \leq h} X(t) > u \right\}. \tag{4}$$

Remark 2.2. The inclusions underlying (4) hold pathwise for any process for which the displayed events are measurable. The first inequality in (4) connects the sliding-window event to a finite-dimensional Gaussian probability. The last inequality shows that both repeated-window probabilities are bounded above by the classical ruin probability over the first reporting period. This yields a sharper bound than the classical ruin probability over the full horizon.

For $m \geq 2$, define

$$G_m(h, c) := \mathbb{P}\left\{ \bigcap_{k=1}^{m-2} \left\{ \sup_{t \in [kh, (k+1)h]} \big(B(t) - ct \big) > 0 \right\} \right\} \tag{5}$$

with the convention $G_2(h, c) := 1$. This is the finite-level crossing probability for the limiting post-hit process. The interval $[0, h]$ imposes no condition in this constant since Brownian motion with drift has positive values arbitrarily close to zero almost surely.

Theorem 2.3. *As $u \to \infty$ we have*

$$p_h^{\mathrm{bl}}(u, T) \sim 2 G_m(h, c) \Psi\left(\frac{u + ch}{\sqrt{h}} \right). \tag{6}$$

For $m \geq 2$, define the sliding constant

$$G_m^{\mathrm{sl}}(h, c) := \mathbb{P}\left\{ \inf_{0 \leq r \leq (m-2)h} \sup_{r \leq s \leq r+h} \big(B(s) - cs \big) > 0 \right\}. \tag{7}$$

For $m = 2$, the infimum in (7) is over the singleton $\{0\}$ and hence $G_2^{\mathrm{sl}}(h, c) = 1$ by the immediate upward oscillations of Brownian motion.
For $m \geq 2$, define the finite-dimensional grid constant

$$J_m(h, c) := \mathbb{P}\left\{ \bigcap_{j=1}^{m-2} \{B(jh) - cjh > 0\} \right\}, \tag{8}$$

with the convention $J_2(h, c) := 1$. For $m \geq 3$, the defining Gaussian vector is non-degenerate and therefore $J_m(h, c) > 0$.

Theorem 2.4. *As $u \to \infty$ we have*

$$p_h^{\mathrm{sl}}(u,T) \sim 2G_m^{\mathrm{sl}}(h,c)\Psi\left(\frac{u+ch}{\sqrt{h}}\right) \tag{9}$$

and, moreover,

$$0 < J_m(h,c) \le G_m^{\mathrm{sl}}(h,c) \le G_m(h,c) \le 1, \qquad m \ge 2. \tag{10}$$

Remark 2.5. The asymptotics in Theorems 2.3 and 2.4 share the classical first-block ruin cost $2\Psi((u+ch)/\sqrt{h})$. Their remaining factors G_m and G_m^{sl} describe the subsequent finite-level requirements; neither probability is a product of one-block ruin probabilities. For $m = 2$, both factors equal one. Conditional on first-block ruin, the first hitting time τ_u of u approaches h. Measured from this hit, the second block is $[d_u, d_u + h]$, where $d_u = h - \tau_u$ tends to zero in conditional probability. Its crossing probability therefore tends to $\mathbb{P}\{\sup_{0\le s\le h}(B(s) - cs) > 0\} = 1$. The sliding requirement has the same limit by right-continuity of its post-hit survival probability; see the proof of Theorem 2.4. Consequently,

$$p_h^{\mathrm{sl}}(u,2h) \sim p_h^{\mathrm{bl}}(u,2h) \sim \psi(u,h).$$

The constant in (7) is a finite-horizon Parisian-type survival probability. If

$$\tau_h^- := \inf\left\{t \ge h : \sup_{t-h\le s\le t} X(s) \le 0\right\},$$

then

$$G_m^{\mathrm{sl}}(h,c) = \mathbb{P}\left\{\tau_h^- > (m-1)h\right\}. \tag{11}$$

3. Extensions and discussion

In this section we discuss three extensions of the Brownian risk model. The first is a compound-Poisson risk process perturbed by Brownian motion, the second is a Brownian model with exponential resetting, and the third is simultaneous block ruin for correlated bivariate Brownian risks.

3.1. Brownian-perturbed compound-Poisson risk model.

In order to incorporate individual claims alongside continuous fluctuations, we consider the classical compound-Poisson risk process perturbed by Brownian motion. We use aggregate-loss notation; see [13, 10]. In this subsection let

$$S(t) = \sum_{j=1}^{N(t)} Y_j, \qquad X(t) = \sigma B(t) + S(t) - ct, \qquad t \ge 0,$$

where N is a Poisson process with rate $\lambda > 0$, the claim sizes Y_j are i.i.d. and non-negative with distribution F, and N is independent of the claims. Assume that B is independent of S, $\sigma \ge 0$, and $c > 0$. The reserve is $R_u(t) = u - X(t)$; the Brownian component allows continuous fluctuations between claim arrivals.

Write $\bar{F}(u) = \mathbb{P}\{Y_1 > u\}$, $u \ge 0$, and assume that F is subexponential; see [11]. Define p_h^{bl} and p_h^{sl} by (1) and (2) using this X. No finite-mean or net-profit condition is needed for the finite fixed-horizon result below.

Theorem 3.1. *For the model above as $u \to \infty$*

$$p_h^{\mathrm{sl}}(u,T) \sim p_h^{\mathrm{bl}}(u,T) \sim \lambda h \bar{F}(u).$$

The same equivalent holds if sliding ruin requires a strict exceedance in every window separately: this event lies between the strict-infimum event in (2) and the block event.

3.2. **Block ruin with exponential resetting.** Let N_ν be a Poisson process of rate $\nu > 0$, independent of B, and let $R_\nu(t)$ denote its last jump time not exceeding t, with $R_\nu(t) = 0$ before the first jump. Put

$$X_\nu(t) = B(t) - B(R_\nu(t)) - c\big(t - R_\nu(t)\big), \qquad t \geq 0. \tag{12}$$

This is the model in [9] with initial value and reset level both at zero. The waiting times between resets are independent exponential variables of rate ν. At each reset the entire drifted loss returns to zero, so the corresponding reserve returns to u. Let $p^{\mathrm{bl}}_{h,\nu}(u,T)$ denote (1) with X replaced by X_ν.

For a fresh Brownian motion write $Y(t) = \widetilde{B}(t) - ct$ and define

$$\theta_x = \inf\{t \geq 0 : x + Y(t) > 0\}, \qquad g_\nu(x;h) = \mathbb{E}\big\{\exp(-\nu\theta_x)\mathbf{1}_{\{\theta_x < h\}}\big\}, \qquad x \in \mathbb{R}, \tag{13}$$

where the integrand is zero when $\theta_x = \infty$. Thus $g_\nu(x;h)$ is the probability of crossing zero before both time h and the first independent exponential reset. In particular, $g_\nu(x;h) = 1$ for $x \geq 0$, including $x = 0$ by immediate upward oscillations. For $x < 0$, with $\kappa_\nu = \sqrt{c^2 + 2\nu}$,

$$g_\nu(x;h) = \exp((c-\kappa_\nu)x)\Psi\left(\frac{-x + \kappa_\nu h}{\sqrt{h}}\right) + \exp((c+\kappa_\nu)x)\Psi\left(\frac{-x - \kappa_\nu h}{\sqrt{h}}\right). \tag{14}$$

The function $g_\nu(\cdot;h)$ is continuous and takes values in $[0,1]$.

Set $a_m = (m-2)h$ and

$$\Gamma^{(\nu)}_m(h,c) = \mathbb{E}\left\{ g_\nu(Y(a_m);h) \prod_{k=1}^{m-3} \mathbf{1}_{\{\sup_{kh \leq t \leq (k+1)h} Y(t) > 0\}} \right\}. \tag{15}$$

The product is empty for $m = 2, 3$, and $\Gamma^{(\nu)}_2(h,c) = g_\nu(0;h) = 1$, $\Gamma^{(\nu)}_3(h,c) = \mathbb{E}\{g_\nu(Y(h);h)\}$. For comparison with the model without resetting, set $g_0(x;h) = \mathbb{P}\{\theta_x < h\}$ and define $\Gamma^{(0)}_m$ by (15) with g_ν replaced by g_0. Conditioning on the start of the final block gives $\Gamma^{(0)}_m(h,c) = G_m(h,c)$. Moreover, $g_\nu \uparrow g_0$ as $\nu \downarrow 0$, so $\Gamma^{(\nu)}_m(h,c) \to G_m(h,c)$.

Theorem 3.2. *For fixed $h, c, \nu > 0$ and $m \geq 2$, as $u \to \infty$,*

$$p^{\mathrm{bl}}_{h,\nu}(u, mh) \sim 2\exp(-\nu(m-1)h)\Gamma^{(\nu)}_m(h,c)\Psi\left(\frac{u+ch}{\sqrt{h}}\right). \tag{16}$$

Moreover, we have

$$0 < \exp(-\nu h)G_m(h,c) \leq \Gamma^{(\nu)}_m(h,c) \leq G_m(h,c). \tag{17}$$

The factor $\exp(-\nu(m-1)h)$ accounts for the absence of resets up to the start of the last block. Within that block, g_ν requires the next crossing to precede the reset; resets after this crossing are harmless. For two blocks, (16) becomes

$$p^{\mathrm{bl}}_{h,\nu}(u, 2h) \sim 2\exp(-\nu h)\Psi\left(\frac{u+ch}{\sqrt{h}}\right), \qquad u \to \infty.$$

For the separate one-block case $m = 1$, write

$$Q_\nu(u,h) = \mathbb{P}\left\{ \sup_{0 \leq t \leq h} X_\nu(t) > u \right\}$$

for classical one-block ruin with resetting. For initial value and reset level zero, [9, Theorem 3(i)] yields

$$Q_\nu(u,h) \sim e^{-\nu h}\psi(u,h) \sim 2e^{-\nu h}\Psi\left(\frac{u+ch}{\sqrt{h}}\right).$$

The first-passage and cycle bounds used below also yield a direct proof. Consequently, the additional factor for repeated block ruin, relative to one-block ruin, is

$$\lim_{u\to\infty}\frac{p^{\mathrm{bl}}_{h,\nu}(u,mh)}{Q_\nu(u,h)}=e^{-\nu(m-2)h}\Gamma^{(\nu)}_m(h,c). \tag{18}$$

For $m=2$ this limit is one. The common factor $e^{-\nu h}$ accounts for avoiding a reset before the first high crossing, which localises near h. The conditional probability of a crossing in the second block before the next reset then tends to one; no additional reset-free interval of length h is required.
For $m=3$, the coefficient is a one-dimensional integral with an explicit integrand:

$$\Gamma^{(\nu)}_3(h,c)=\Psi(c\sqrt{h})+\frac{1}{\sqrt{h}}\int_{-\infty}^{0}g_\nu(x;h)\varphi\left(\frac{x+ch}{\sqrt{h}}\right)dx. \tag{19}$$

3.3. **Simultaneous block ruin.** Let $\rho\in(-1,1)$, let B_1 and B_2 be independent copies of B, and put

$$W_1(t)=B_1(t),\qquad W_2(t)=\rho B_1(t)+\sqrt{1-\rho^2}B_2(t),\qquad t\ge 0. \tag{20}$$

Fix $c_1,c_2>0$ and $a\in(\rho,1]$, and define

$$X_i(t)=W_i(t)-c_it,\qquad i=1,2.$$

Set

$$p^{\mathrm{sbl}}_{h,\rho}(u,T):=\mathbb{P}\left\{\bigcap_{k=1}^{m}\{\exists t\in[(k-1)h,kh]:X_1(t)>u,\ X_2(t)>au\}\right\}. \tag{21}$$

Thus (21) requires simultaneous ruin at least once in every reporting block. The insurance interpretation with two positive initial reserves corresponds to $a>0$; the mathematical result below also covers $a\le 0$ when $\rho<a$.
Let

$$\varphi_\rho(x,y):=\frac{1}{2\pi\sqrt{1-\rho^2}}\exp\left(-\frac{x^2-2\rho xy+y^2}{2(1-\rho^2)}\right)$$

be the density of $(W_1(1),W_2(1))$. Define

$$\lambda_1=\frac{1-a\rho}{1-\rho^2},\qquad \lambda_2=\frac{a-\rho}{1-\rho^2}, \tag{22}$$

and the simultaneous-ruin constant

$$C_{a,\rho}:=\int_{\mathbb{R}^2}\mathbb{P}\{\exists t\ge 0:W_1(t)-t>x,\ W_2(t)-at>y\}\exp(\lambda_1x+\lambda_2y)\,dx\,dy. \tag{23}$$

This is the constant appearing in [8]. Finally, for $m\ge 3$, set

$$\mathcal{H}_m(h,c_1,c_2,\rho):=\mathbb{P}\left\{\bigcap_{k=1}^{m-2}\{\exists t\in[kh,(k+1)h]:W_1(t)-c_1t>0,\ W_2(t)-c_2t>0\}\right\} \tag{24}$$

and set $\mathcal{H}_2(h,c_1,c_2,\rho)=1$. For $m\ge 3$, requiring both coordinates to be positive at each time $h,\dots,(m-2)h$ yields a positive-probability Gaussian grid event contained in (24) and hence $\mathcal{H}_m(h,c_1,c_2,\rho)>0$.

Theorem 3.3. *For the model above as $u\to\infty$*

$$p^{\mathrm{sbl}}_{h,\rho}(u,T)\sim\mathcal{H}_m(h,c_1,c_2,\rho)C_{a,\rho}\frac{h}{u^2}\varphi_\rho\left(\frac{u+c_1h}{\sqrt{h}},\frac{au+c_2h}{\sqrt{h}}\right). \tag{25}$$

The regime $a \le \rho$, in which the first-block asymptotics exhibit dimension reduction as in [8], is omitted here.

3.4. **Discussion.** The scalar Brownian proofs share a first-hitting-time argument. Conditional on a first-block exceedance, the first hit of u approaches h; the strong Markov property then separates the first-block ruin cost from the remaining finite-level event. For blocks this event has probability $G_m(h,c)$, whereas sliding windows yield the Parisian-type survival probability $G_m^{\mathrm{sl}}(h,c)$. Thus the full horizon enters through these constants. In particular, the behaviour remains that of one-point asymptotics up to a multiplicative constant. Indeed, for $X(t) = B(t) - ct$,

$$\mathbb{P}\{X(h) > u\} = \Psi\left(\frac{u+ch}{\sqrt{h}}\right),$$

and hence

$$p_h^{\mathrm{bl}}(u,T) \sim 2G_m(h,c)\mathbb{P}\{X(h) > u\}, \quad p_h^{\mathrm{sl}}(u,T) \sim 2G_m^{\mathrm{sl}}(h,c)\mathbb{P}\{X(h) > u\}. \tag{26}$$

The later monitoring requirements change the multiplicative constant but leave the one-point tail order at time h unchanged.

In the perturbed model the subexponential tail of the total claims in the first block determines the asymptotics. Once these claims keep the process above u throughout $[h,T]$, every length-h window is covered. Bounds using the Brownian extrema show that the perturbation changes neither the order nor the constant λh.

With exponential resetting, the leading contribution comes from the initial Brownian path reaching every reporting block before its first reset. A high crossing arising after a reset in the first block is negligible at the first-block scale. The same holds for a second high excursion after an initial crossing. The final crossing can occur before T, which is why requiring no reset throughout $[0,T]$ would yield only a lower bound.

For simultaneous bivariate ruin, conditioning on the endpoint at time h and an exact Brownian-bridge time change yield the same factorisation into a first-block cost and the later-block factor $\mathcal{H}_m(h,c_1,c_2,\rho)$. Here too the behaviour is that of one-point asymptotics up to a multiplicative constant. Since $\lambda_1, \lambda_2 > 0$ in the binding regime $\rho < a \le 1$, the standard Gaussian vector tail asymptotics in [15, Theorem 4.1] yield

$$\mathbb{P}\{X_1(h) > u,\ X_2(h) > au\} \sim \frac{h}{\lambda_1\lambda_2 u^2}\varphi_\rho\left(\frac{u+c_1h}{\sqrt{h}}, \frac{au+c_2h}{\sqrt{h}}\right). \tag{27}$$

Consequently, (25) is equivalent to

$$p_{h,\rho}^{\mathrm{sbl}}(u,T) \sim \lambda_1\lambda_2 C_{a,\rho}\mathcal{H}_m(h,c_1,c_2,\rho)\mathbb{P}\{X_1(h) > u,\ X_2(h) > au\}. \tag{28}$$

Thus the repeated simultaneous-ruin probability retains the one-point Gaussian tail order, with the later blocks affecting only the multiplicative constant.

Across these models dependence across windows is essential: the repeated-ruin probability is not a product of one-window probabilities.

4. Numerical examples

Throughout this section we take $h = 1$ and $c = 0.3$; in Section 4.2 the reset rate is $\nu = 0.4$. To simulate crossings between observation times, we use the Brownian-bridge method; see, e.g., [14]. Let x and y be the values of a Brownian motion with drift at two consecutive simulation

points $t_0 < t_1$ and put $\Delta t = t_1 - t_0$. Conditional on these values, the probability that the level u is exceeded on $[t_0, t_1]$ equals one if $x > u$ or $y > u$, and otherwise

$$\pi_{\Delta t}(x, y) = \exp\left\{-\frac{2(u-x)(u-y)}{\Delta t}\right\};$$

the drift does not enter. With an independent $U \sim \text{Uniform}(0,1)$, the indicator

$$\chi = \begin{cases} 1, & \text{if } x > u \text{ or } y > u, \\ \mathbf{1}_{\{U < \pi_{\Delta t}(x,y)\}}, & \text{if } x \le u \text{ and } y \le u, \end{cases}$$

has exactly the conditional law of the crossing event. Conditional on the simulation values, the Brownian bridges on disjoint subintervals are independent, so their crossing indicators are generated using independent uniforms. The same indicator is reused whenever its subinterval belongs to more than one monitoring window. This preserves the dependence between overlapping windows, and events determined by these indicators are simulated without time-discretisation bias.

For an event of probability p, let $I_1, \dots, I_{N_{\text{rep}}}$ be its indicators in N_{rep} independent replications. We report the Monte Carlo estimate and its estimated standard error,

$$\widehat{p} = \frac{1}{N_{\text{rep}}} \sum_{\ell=1}^{N_{\text{rep}}} I_\ell, \qquad \widehat{\text{se}}(\widehat{p}) = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{N_{\text{rep}}}}.$$

For sliding windows, the two grid events defined below provide lower and upper bounds on the desired probability.

4.1. **Block and sliding ruin for the Brownian model.** We first illustrate Theorems 2.3 and 2.4 for $m = 3, 4$. Each of the m reporting blocks is divided into n simulation cells, giving the grid $t_j = j\delta$, $j = 0, \dots, mn$, with $\delta = h/n$ and $n \ge 3$. For $X(t) = B(t) - ct$, let χ_i denote the crossing indicator of $[t_i, t_{i+1}]$, generated by the Brownian-bridge method. The indicator of a crossing in reporting block k is

$$\beta_k = \max_{(k-1)n \le i < kn} \chi_i, \qquad k = 1, \dots, m,$$

so $\prod_{k=1}^m \beta_k$ has exactly the law of the block-ruin indicator in (1). The reporting-block endpoints alone suffice for this calculation; the finer grid also permits bounds for sliding windows.

For $r \in [t_j, t_{j+1}]$, $j = 0, \dots, (m-1)n - 1$, we have

$$[t_{j+1}, t_{j+n}] \subseteq [r, r+h] \subseteq [t_j, t_{j+n+1}].$$

A crossing on every inner interval implies the sliding event: the finitely many inner maxima then exceed u by a common positive margin. Conversely, the sliding event implies a crossing on every outer interval. These two grid events are determined, respectively, by

$$\max_{j+1 \le i < j+n} \chi_i = 1 \quad \text{and} \quad \max_{j \le i \le j+n} \chi_i = 1 \qquad \text{for every } j.$$

Their probabilities therefore bracket $p_h^{\text{sl}}(u, T)$, and both events are simulated without time-discretisation bias. The reported bracket endpoints are Monte Carlo estimates of these probabilities; they remain subject to sampling error and are not confidence limits.

The constants $G_m(h, c)$ and $G_m^{\text{sl}}(h, c)$ are estimated by applying the same method to their defining events, using paths of $Y(t) = \widetilde{B}(t) - ct$ on $[0, (m-1)h]$ at level zero, where $\widetilde{B}$ is a

standard Brownian motion. Conditioning on $Y(h)$ yields the integral representation

$$G_3(h,c) = \Psi(c\sqrt{h}) + \frac{1}{\sqrt{h}} \int_{-\infty}^{0} \psi(-x,h)\varphi\left(\frac{x+ch}{\sqrt{h}}\right) dx,$$

which serves as a check, and $J_m(h,c)$ is a Gaussian orthant probability. Table 1 reports the constants for $m = 3, 4, 5$ ($2 \cdot 10^6$ replications, $\delta = h/200$). The midpoint estimates of the sliding constants are approximately 10%, 14% and 16% below the respective block constants. The grid lower bound J_m in (10) is substantially smaller in all three cases.

m	$J_m(h,c)$	$G_m^{\mathrm{sl}}(h,c)$	$G_m(h,c)$
3	0.3821	$[0.5534, 0.5567]$	0.6182
4	0.2408	$[0.3960, 0.3988]$	0.4641
5	0.1740	$[0.3039, 0.3063]$	0.3643

TABLE 1. The constants in (10) for $h = 1$, $c = 0.3$. The interval for G_m^{sl} reports estimates of the two bracketing grid-event probabilities; the Monte Carlo standard errors are below $4 \cdot 10^{-4}$. Numerical integration yields $G_3(h,c) = 0.618185$.

Table 2 compares the block estimates and the estimated sliding brackets with the asymptotic approximations in Theorems 2.3 and 2.4. We use the numerical estimates of G_m and the midpoints of the brackets for G_m^{sl} in Table 1. Each row uses $4 \cdot 10^6$ replications and $\delta = h/50$; the relative standard errors are at most about 1.2% for $u \le 2.5$ and 2.7% for $u = 3$. The block ratios lie between 0.98 and 1.03. Each sliding-ratio range is obtained by dividing both estimated bracket endpoints by the same numerical asymptotic approximation. These ranges contain one except when $m = 4$ and $u = 1$, where the range is $[1.012, 1.079]$.

m	u	$\widehat{p}_h^{\mathrm{bl}}$	$\widehat{p}_h^{\mathrm{sl}}$ (bracket)	$2G_m\Psi$	$2G_m^{\mathrm{sl}}\Psi$	block ratio	sliding ratio
3	1.0	0.119687	$[0.104972, 0.111659]$	0.119681	0.107449	1.000	$[0.977, 1.039]$
3	1.5	0.044749	$[0.039054, 0.042750]$	0.044423	0.039883	1.007	$[0.979, 1.072]$
3	2.0	0.013416	$[0.011555, 0.013132]$	0.013259	0.011904	1.012	$[0.971, 1.103]$
3	2.5	0.003176	$[0.002666, 0.003184]$	0.003159	0.002836	1.005	$[0.940, 1.123]$
3	3.0	0.000584	$[0.000474, 0.000603]$	0.000598	0.000537	0.977	$[0.883, 1.124]$
4	1.0	0.092185	$[0.077879, 0.083022]$	0.089850	0.076937	1.026	$[1.012, 1.079]$
4	1.5	0.034063	$[0.028516, 0.031274]$	0.033351	0.028557	1.021	$[0.999, 1.095]$
4	2.0	0.010250	$[0.008453, 0.009620]$	0.009954	0.008524	1.030	$[0.992, 1.129]$
4	2.5	0.002406	$[0.001953, 0.002325]$	0.002372	0.002031	1.014	$[0.962, 1.145]$
4	3.0	0.000441	$[0.000346, 0.000433]$	0.000449	0.000384	0.983	$[0.901, 1.127]$

TABLE 2. Block and sliding ruin for the Brownian model, $h = 1$, $c = 0.3$, $4 \cdot 10^6$ replications per row. The asymptotic approximations use $G_3 = 0.6182$, $G_3^{\mathrm{sl}} = 0.5550$, $G_4 = 0.4641$ and $G_4^{\mathrm{sl}} = 0.3974$; the two sliding constants are approximated by the midpoints of their brackets in Table 1. The sliding-ratio ranges use these fixed midpoint approximations; uncertainty in the estimated constants is not propagated.

The agreement observed at these moderate levels depends on the premium rate and the number of blocks. For sliding ruin, its precision is also limited by the width of the grid

brackets. By (3) and Mills' ratio, the first-block ruin probability satisfies

$$\psi(u,h)=2\Psi\left(\frac{u+ch}{\sqrt{h}}\right)\left(1+\frac{ch}{u}+O(u^{-2})\right).$$

Writing τ_u for the first hitting time of u, the same formula yields, for every $x\geq 0$,

$$\mathbb{P}\left\{\frac{u^2(h-\tau_u)}{2h^2}>x\mid \tau_u<h\right\}=\frac{\psi(u,h-2h^2x/u^2)}{\psi(u,h)}\longrightarrow e^{-x},\qquad u\to\infty,$$

where the ratio is understood for sufficiently large u. Thus the post-hit kernels in Section 5 are sampled at distances of order u^{-2} from zero. Their continuity yields the first-order asymptotics, but a rate for the full approximation also requires a quantitative estimate of their variation near zero. We do not derive such an estimate here. For $m=2$, where $p_h^{\mathrm{bl}}(u,2h)=\mathbb{E}\{\mathbf{1}_{\{\tau_u<h\}}K_2(h-\tau_u)\}$ can be evaluated by numerical integration, the ratio $p_h^{\mathrm{bl}}(u,2h)/(2\Psi((u+ch)/\sqrt{h}))$ equals 0.87, 0.90, 0.94 and 0.97 for $u=1,2,4,8$. For $c=1$ and $m=3$, the block ratio of Table 2 is about 1.27 at $u=1$ and 1.16 at $u=2$.

4.2. **Block ruin with exponential resetting.** We now consider the model of Section 3.2 with $\nu=0.4$ and $m=2,3$, for which $\Gamma_2^{(\nu)}(h,c)=1$, while numerical evaluation of (19) yields

$$\Gamma_3^{(\nu)}(h,c)=0.595270.$$

For large negative x, the first term in (14) combines a large exponential with a small normal tail. We evaluate the terms on the logarithmic scale.

To estimate $p_{h,\nu}^{\mathrm{bl}}(u,mh)$, we first generate the reset epochs of the Poisson process with rate ν on $[0,mh]$, merge them with the block boundaries $0,h,\dots,mh$, and sort the resulting sequence. On each interval $[t_0,t_1]$ between consecutive points, the process is a Brownian motion with drift $-c$ without resetting. Conditional on $X_\nu(t_0)=x$, the value just before t_1 is simulated as

$$X_\nu(t_1-)=x-c\Delta t+\sqrt{\Delta t}\xi,\qquad \Delta t=t_1-t_0,\quad \xi\sim N(0,1),$$

and the crossing indicator χ of $[t_0,t_1]$ is computed from x and $X_\nu(t_1-)$. If t_1 is a reset epoch, the process is then restarted at zero, $X_\nu(t_1)=0$, in accordance with (12). For each block $[(k-1)h,kh]$, the block indicator β_k is the maximum of χ over the subintervals contained in the block; a value above u at a shared boundary kh counts for both adjacent blocks, as the blocks are closed. The block ruin indicator of a replication is $\prod_{k=1}^m\beta_k$, and $\widehat{p}$ denotes its average over $N_{\mathrm{rep}}=2\cdot 10^7$ replications.

We compare $\widehat{p}$ not only with the asymptotic approximation (16), but also with the main term of its proof. The event $M_u\subseteq E_u$ defined in the proof of Theorem 3.2 requires a crossing in every block before the first reset and utilising (40)

$$\mathbb{P}\{M_u\}=e^{-\nu(m-1)h}\int_0^h K_{m,\nu}(h-t)\frac{u}{\sqrt{2\pi t^3}}\exp\left(-\frac{(u+ct)^2}{2t}\right)dt,$$

with the first-passage density of Remark 5.2. Both kernels needed here admit one-dimensional representations. For $d>0$ we have

$$K_{2,\nu}(d)=\Psi(c\sqrt{d})+\frac{1}{\sqrt{d}}\int_{-\infty}^0 g_\nu(y;h)\varphi\left(\frac{y+cd}{\sqrt{d}}\right)dy.$$

At $d=0$ this kernel equals one. For $0\leq d\leq h$ setting $s=d+h$, $\Phi=1-\Psi$ we have

$$K_{3,\nu}(d)=\Psi(c\sqrt{s})+\frac{2}{\sqrt{s}}\int_{-\infty}^0 g_\nu(y;h)\Phi\left(y\sqrt{\frac{d}{hs}}\right)\varphi\left(\frac{y+cs}{\sqrt{s}}\right)dy. \tag{29}$$

At $d = 0$ immediate upward oscillations make the crossing condition automatic, and the formula reduces to (19) because $2\Phi(0) = 1$.
Since $M_u \subseteq E_u$, the probability $\mathbb{P}\{M_u\}$ is an exact lower bound for $p^{\mathrm{bl}}_{h,\nu}(u, mh)$; the difference is the remainder $\mathbb{P}\{E_u \setminus M_u\}$ shown to be negligible in the proof. Table 3 displays the Monte Carlo estimates, their standard errors, numerical values of this main term, the asymptotic approximation and the two ratios.

m	u	$\widehat{p}$	$\widehat{\mathrm{se}}$	$\mathbb{P}\{M_u\}$	asymptotic	$\widehat{p}$/asympt.	$\widehat{p}/\mathbb{P}\{M_u\}$
2	1.0	0.144866	$7.9 \cdot 10^{-5}$	0.108758	0.129775	1.116	1.332
2	1.5	0.049042	$4.8 \cdot 10^{-5}$	0.041390	0.048170	1.018	1.185
2	2.0	0.014143	$2.6 \cdot 10^{-5}$	0.012599	0.014377	0.984	1.123
2	2.5	0.003320	$1.3 \cdot 10^{-5}$	0.003049	0.003426	0.969	1.089
2	3.0	0.000613	$5.5 \cdot 10^{-6}$	0.000584	0.000648	0.945	1.048
2	3.5	$9.13 \cdot 10^{-5}$	$2.1 \cdot 10^{-6}$	$8.84 \cdot 10^{-5}$	$9.70 \cdot 10^{-5}$	0.942	1.033
2	4.0	$1.20 \cdot 10^{-5}$	$7.8 \cdot 10^{-7}$	$1.05 \cdot 10^{-5}$	$1.14 \cdot 10^{-5}$	1.05	1.14
3	1.0	0.083450	$6.2 \cdot 10^{-5}$	0.052251	0.051783	1.612	1.597
3	1.5	0.025034	$3.5 \cdot 10^{-5}$	0.019456	0.019221	1.302	1.287
3	2.0	0.006723	$1.8 \cdot 10^{-5}$	0.005817	0.005737	1.172	1.156
3	2.5	0.001509	$8.7 \cdot 10^{-6}$	0.001387	0.001367	1.104	1.088
3	3.0	0.000282	$3.8 \cdot 10^{-6}$	0.000263	0.000259	1.091	1.074
3	3.5	$4.20 \cdot 10^{-5}$	$1.5 \cdot 10^{-6}$	$3.93 \cdot 10^{-5}$	$3.87 \cdot 10^{-5}$	1.085	1.069
3	4.0	$5.0 \cdot 10^{-6}$	$5.0 \cdot 10^{-7}$	$4.6 \cdot 10^{-6}$	$4.6 \cdot 10^{-6}$	1.11	1.09

TABLE 3. Block ruin with resetting, $h = 1$, $c = 0.3$, $\nu = 0.4$, $2 \cdot 10^7$ replications per row. $\mathbb{P}\{M_u\}$ is the main term of the proof of Theorem 3.2, evaluated by quadrature; the asymptotic approximation is (16).

Three features of Table 3 deserve comment. First, the ratio of $\widehat{p}$ to the asymptotic approximation is not close to one at these levels: for $m = 2$ it decreases from 1.12 at $u = 1$ to 0.94 at $u = 3.5$, and for $m = 3$ from 1.61 to 1.09. Equivalently, the implied values $\widehat{\Gamma}_3 = \widehat{p}/(2e^{-2\nu h}\Psi((u + ch)/\sqrt{h}))$ decrease from 0.96 to 0.65, against the limit 0.595. Second, the main term alone behaves as in Section 4.1: the ratio of $\mathbb{P}\{M_u\}$ to the asymptotic approximation increases from 0.84 to 0.92 for $m = 2$ and equals about 1.01 for $m = 3$. These values combine the first-block approximation error with the variation of the post-hit kernel. Third, the remainder $\mathbb{P}\{E_u \setminus M_u\}$, relative to the asymptotic approximation, is estimated by the difference between $\widehat{p}$ and $\mathbb{P}\{M_u\}$, divided by that approximation. The estimates are 0.28, 0.16, 0.11, 0.08, 0.04, 0.03 for $m = 2$ and 0.60, 0.29, 0.16, 0.09, 0.08, 0.07 for $m = 3$ at $u = 1, 1.5, \ldots, 3.5$. This remainder consists of paths with a reset followed by a new crossing. The bounds $\nu \int_0^h \psi(u, h - s)\, ds$ and $(1 + \nu T)\psi(u, h)\psi(u, T)$ in the proof are of relative order u^{-2} and $\psi(u, T)$, respectively. The first assertion follows from $\int_0^h \psi(u, h - s)\, ds = \mathbb{E}\{(h - \tau_u)\mathbf{1}_{\{\tau_u < h\}}\}$ and the first-passage density. These bounds prove negligibility as $u \to \infty$, but need not be sharp at the moderate levels in the table. The last column shows that $\widehat{p}/\mathbb{P}\{M_u\}$ decreases towards one, from 1.33 to 1.03 for $m = 2$ and from 1.60 to 1.07 for $m = 3$; the entries for $u = 4$ rest on 241 and 101 events and have relative standard errors of 6% and 10%.
Finally, Table 4 shows the dependence of $\Gamma_3^{(\nu)}(h, c)$ on the reset rate, obtained from (19), together with the lower bound in (17); the upper bound is $G_3(h, c) = 0.6182$ for every ν, and $\Gamma_3^{(\nu)}(h, c) \to G_3(h, c)$ as $\nu \downarrow 0$. The constant decreases slowly with ν, since a reset only

matters if it precedes the final crossing, whereas the lower bound $e^{-\nu h}G_3(h,c)$ charges a full reset-free block.

ν	0	0.1	0.2	0.4	0.8	1.6	3.2
$e^{-\nu h}G_3(h,c)$	0.6182	0.5594	0.5061	0.4144	0.2778	0.1248	0.0252
$\Gamma_3^{(\nu)}(h,c)$	0.6182	0.6120	0.6061	0.5953	0.5767	0.5487	0.5144

TABLE 4. $\Gamma_3^{(\nu)}(h,c)$ for $h=1$, $c=0.3$ as a function of ν, with the lower bound of (17).

5. Proofs

We first establish first-passage localisation. We use the usual augmented natural filtrations, including the independent reset process when present. On $C([0,T])$ with the uniform norm, window suprema are jointly continuous in the path and the interval endpoints. The infimum of the length-h window suprema is likewise jointly continuous in the path and the horizon. Consequently, the post-hit indicators and payoffs in (32), (34) and (40) are jointly Borel measurable in the path and the shift or horizon, as required for conditioning at the first hitting time.

For the scalar Brownian proofs let $X(t)=B(t)-ct$ with $c>0$. Define the first hitting time

$$\tau_u := \inf\{t\ge 0 : X(t)=u\}.$$

In expectations below, products containing $\mathbf{1}_{\{\tau_u<h\}}$ are understood to be zero on $\{\tau_u\ge h\}$. Immediate upward oscillations after τ_u imply that $\{\tau_u<h\}=\{\sup_{0\le t\le h}X(t)>u\}$ almost surely.

Lemma 5.1. *As $u\to\infty$ we have*

$$\mathbb{P}\{\tau_u<h\}\sim 2\Psi\left(\frac{u+ch}{\sqrt{h}}\right), \qquad \tau_u \mid \{\tau_u<h\} \xrightarrow[u\to\infty]{} h \quad \text{in probability.} \tag{30}$$

Consequently, if $q:(0,h]\to\mathbb{R}$ is bounded and measurable and $q(d)\to q_0$ as $d\downarrow 0$, then

$$\frac{\mathbb{E}\{\mathbf{1}_{\{\tau_u<h\}}q(h-\tau_u)\}}{\mathbb{P}\{\tau_u<h\}} \xrightarrow[u\to\infty]{} q_0. \tag{31}$$

Proof. The first assertion follows from (3) and Mills' ratio. For every $\delta\in(0,h)$ the same formula yields

$$\mathbb{P}\{\tau_u\le h-\delta \mid \tau_u<h\}=\frac{\psi(u,h-\delta)}{\psi(u,h)}\xrightarrow[u\to\infty]{}0,$$

because the numerator has a strictly larger Gaussian exponential cost. This proves the conditional convergence. Boundedness of q and its limit at zero then yield (31). □

Proof of Lemma 2.1. On the grid event in (4), the finite minimum

$$\eta := \min_{1\le k\le m-1}(X(kh)-u)$$

is strictly positive. Every interval of length h in $[0,T]$ contains one of the points $h,2h,\dots,(m-1)h$. Its supremum is therefore at least $u+\eta$. This proves the first inequality. Each reporting block is a particular sliding window. This proves the second inequality; retaining only the first block proves the last one. □

Proof of Theorem 2.3. Let $Y(t) = \widetilde{B}(t) - ct$, where $\widetilde{B}$ is a standard Brownian motion, and define, for $0 \le d \le h$,

$$K_m(d) := \mathbb{P}\left\{ \bigcap_{k=0}^{m-2} \left\{ \sup_{d+kh \le s \le d+(k+1)h} Y(s) > 0 \right\} \right\}.$$

On $\{\tau_u < h\}$, the first block contains an exceedance almost surely. The strong Markov property at τ_u therefore yields the exact identity

$$p_h^{\mathrm{bl}}(u,T) = \mathbb{E}\{\mathbf{1}_{\{\tau_u<h\}} K_m(h-\tau_u)\}. \tag{32}$$

Indeed, relative to τ_u, the remaining reporting blocks are $[d+kh, d+(k+1)h]$, $k = 0, \dots, m-2$, where $d = h - \tau_u$.
We claim that

$$K_m(d) \underset{d\downarrow 0}{\longrightarrow} K_m(0) = G_m(h,c). \tag{33}$$

For each $k = 0, \dots, m-2$ path continuity implies

$$\sup_{d+kh \le s \le d+(k+1)h} Y(s) \underset{d\downarrow 0}{\longrightarrow} M_k := \sup_{kh \le s \le (k+1)h} Y(s).$$

For $k = 0$, immediate upward oscillations yield $M_0 > 0$ almost surely. For $k = 1, \dots, m-2$ write

$$M_k = Y(kh) + \sup_{0 \le s \le h} \{Y(kh+s) - Y(kh)\}.$$

The two terms are independent, and $Y(kh)$ has a Gaussian density. Thus $\mathbb{P}\{M_k = 0\} = 0$. The indicators of the finite intersections converge almost surely, proving (33) by dominated convergence. The equality $K_m(0) = G_m(h,c)$ follows by discarding the almost sure condition $M_0 > 0$. The same argument at $d > 0$ proves continuity of K_m on $(0,h]$, hence measurability. Apply Lemma 5.1 with $q = K_m$ to (32). Equations (30) and (33) yield

$$p_h^{\mathrm{bl}}(u,T) \sim 2G_m(h,c)\Psi\left(\frac{u+ch}{\sqrt{h}}\right).$$

Here $G_m(h,c) \ge J_m(h,c) > 0$ since the grid event defining $J_m(h,c)$ implies every required block crossing. □

Proof of Theorem 2.4. For $R \ge h$, put

$$Q_{R,h} := \inf_{0 \le r \le R-h} \sup_{r \le s \le r+h} Y(s), \qquad H(R) := \mathbb{P}\{Q_{R,h} > 0\},$$

where $Y(t) = \widetilde{B}(t) - ct$. Since X and Y have the same law, continuity of the moving-window supremum and compactness yield $H(R) = \mathbb{P}\{\tau_h^- > R\}$. In particular

$$H(T-h) = G_m^{\mathrm{sl}}(h,c).$$

As a survival function, H is right-continuous.
We first establish the exact identity

$$p_h^{\mathrm{sl}}(u,T) = \mathbb{E}\{\mathbf{1}_{\{\tau_u<h\}} H(T-\tau_u)\}. \tag{34}$$

The sliding event requires $\tau_u < h$. On this event, use the strong Markov property to write $X(\tau_u + s) - u = Y(s)$. The windows starting at or after τ_u are covered with a uniform positive margin precisely when $Q_{T-\tau_u,h} > 0$. Almost surely, immediate upward oscillations also yield

some $s_0 \in (0, h-\tau_u)$ with $Y(s_0) > 0$. The point $\tau_u + s_0$ lies in $[\tau_u, h]$, the common intersection of all windows $[t, t+h]$ with $0 \le t \le \tau_u$. Hence, when $Q_{T-\tau_u,h} > 0$,

$$\inf_{0\le t\le T-h} \sup_{t\le s\le t+h} X(s) - u \ge \min(Y(s_0), Q_{T-\tau_u,h}) > 0.$$

This proves both directions of the event equivalence and then (34) by conditional expectation. The horizon $T-h+d$ approaches $T-h$ from above. Right-continuity therefore yields

$$H(T-h+d) \underset{d\downarrow 0}{\longrightarrow} H(T-h) = G_m^{\mathrm{sl}}(h,c).$$

Apply Lemma 5.1 with $q(d) = H(T-h+d)$ to obtain

$$\frac{p_h^{\mathrm{sl}}(u,T)}{\Psi((u+ch)/\sqrt{h})} \underset{u\to\infty}{\longrightarrow} 2G_m^{\mathrm{sl}}(h,c). \tag{35}$$

For the comparison of the constants, all three equal one when $m = 2$. If $m \ge 3$, on the event defining $J_m(h,c)$, the finite minimum

$$\eta := \min_{1\le j\le m-2}(Y(jh))$$

is positive. Every length-h window in $[0,(m-1)h]$ contains one of these grid points. Hence $Q_{(m-1)h,h} \ge \eta > 0$. Thus $J_m(h,c) \le G_m^{\mathrm{sl}}(h,c)$. Evaluating the sliding condition at $r = kh$, $k = 1,\dots,m-2$, yields $G_m^{\mathrm{sl}}(h,c) \le G_m(h,c) \le 1$. Together with $J_m(h,c) > 0$ this proves (10) and makes (35) the claimed asymptotic equivalence. □

Remark 5.2. Differentiating (3) with respect to the horizon yields the first-passage density

$$\mathbb{P}\{\tau_u \in dt\} = \frac{u}{\sqrt{2\pi t^3}} \exp\left(-\frac{(u+ct)^2}{2t}\right) dt, \qquad t > 0.$$

Consequently, (34) also yields

$$p_h^{\mathrm{sl}}(u,T) = \int_0^h H(T-t)\frac{u}{\sqrt{2\pi t^3}} \exp\left(-\frac{(u+ct)^2}{2t}\right) dt. \tag{36}$$

Proof of Theorem 3.1. Since the claims are non-negative

$$\mathbb{P}\{S(h) - Z_T > u\} \le p_h^{\mathrm{sl}}(u,T) \le p_h^{\mathrm{bl}}(u,T) \le \mathbb{P}\{S(h) + Z_T > u\}, \tag{37}$$

where $Z_T := cT + \sigma \sup_{0\le s\le T} |B(s)|$. Indeed, on the event on the left, the margin $\eta = S(h) - Z_T - u$ is positive and $X(s) \ge u + \eta$ for every $s \in [h,T]$. Every length-h window meets $[h,T]$. Its supremum is therefore at least $u+\eta$; this also verifies the strict sliding infimum. The middle inequality follows by selecting the reporting blocks. For the last inequality, $X(s) \le S(h) + Z_T$ on $[0,h]$.

Since the Poisson count has finite exponential moments, the random-sum tail asymptotic in [12, Theorem 3.37] yields

$$\mathbb{P}\{S(h) > u\} \sim \lambda h \bar{F}(u), \qquad u \to \infty. \tag{38}$$

The random variable Z_T is independent of $S(h)$ and has finite exponential moments. Since F is long-tailed,

$$\mathbb{P}\{\pm Z_T > u\} = o(\bar{F}(u))$$

for either choice of sign. The convolution result in [12, Corollary 3.19] permits distributions on the whole real line and zero limiting tail ratios. Apply it with reference distribution F and summands $S(h)$ and $\pm Z_T$. Their respective limiting tail ratios are λh and zero. Hence

$$\mathbb{P}\{S(h) \pm Z_T > u\} \sim \lambda h \bar{F}(u), \qquad u \to \infty.$$

The sandwich (37) completes the proof. □

Proof of Theorem 3.2. First, for $x < 0$, the first-passage density yields

$$g_\nu(x;h) = \int_0^h \exp(-\nu t)\frac{-x}{\sqrt{2\pi t^3}}\exp\left(-\frac{(-x+ct)^2}{2t}\right)\,dt.$$

With $\kappa_\nu = \sqrt{c^2+2\nu}$, the integrand is $\exp((c-\kappa_\nu)x)$ times the first-passage density of level $-x$ for Brownian motion with drift $-\kappa_\nu$. Formula (3) therefore yields (14). Its limit as $x \uparrow 0$ is $\Psi(\kappa_\nu\sqrt{h}) + \Psi(-\kappa_\nu\sqrt{h}) = 1$, proving continuity also at zero.

Let $X(t) = B(t) - ct$ denote the unreset process and let ζ_1 be the first reset time. Put $L = (m-1)h$. Write E_u for the block event of X_ν, and let $M_u \subseteq E_u$ require a crossing in every reporting block before ζ_1. In particular, M_u requires $\zeta_1 > L$. Define the unreset event

$$A_{m-1}(u) = \bigcap_{k=1}^{m-1}\left\{\sup_{(k-1)h\le t\le kh} X(t) > u\right\}.$$

On $\{\zeta_1 > L\}$, the first $m-1$ blocks coincide with those of the unreset process. Conditional on the Brownian path up to L and on $\zeta_1 > L$, independent increments and exponential memorylessness yield $g_\nu(X(L)-u;h)$ for a final-block crossing before ζ_1. Since $\mathbb{P}\{\zeta_1 > L\} = e^{-\nu L}$, we obtain

$$\mathbb{P}\{M_u\} = \exp(-\nu L)\mathbb{E}\big\{\mathbf{1}_{\{A_{m-1}(u)\}}g_\nu(X(L)-u;h)\big\}. \tag{39}$$

This identity allows resets after the final required crossing.

For $0 \le d \le h$, set

$$K_{m,\nu}(d) = \mathbb{E}\left\{g_\nu(Y(d+a_m);h)\prod_{k=0}^{m-3}\mathbf{1}_{\{\sup_{d+kh\le t\le d+(k+1)h}Y(t)>0\}}\right\},$$

with an empty product for $m = 2$. The strong Markov property at $\tau_u = \inf\{t \ge 0 : X(t) = u\}$ turns (39) into

$$\mathbb{P}\{M_u\} = \exp(-\nu L)\mathbb{E}\big\{\mathbf{1}_{\{\tau_u<h\}}K_{m,\nu}(h-\tau_u)\big\}. \tag{40}$$

The continuity and no-atom arguments in the proof of Theorem 2.3 and continuity of g_ν imply

$$K_{m,\nu}(d) \underset{d\downarrow 0}{\longrightarrow} \Gamma_m^{(\nu)}(h,c).$$

Indeed, the $k = 0$ crossing condition has probability one at $d = 0$, and the other block maxima have no atom at zero. For $m = 2$, only $g_\nu(Y(d);h)$ remains and converges almost surely to one. The same argument yields continuity of $K_{m,\nu}$ on $(0,h]$, and therefore measurability. Bounded convergence and Lemma 5.1 now yield

$$\frac{\mathbb{P}\{M_u\}}{\psi(u,h)} \underset{u\to\infty}{\longrightarrow} \exp(-\nu L)\Gamma_m^{(\nu)}(h,c). \tag{41}$$

We next show that $\mathbb{P}\{E_u \setminus M_u\} = o(\psi(u,h))$. On $E_u \setminus M_u$, either a cycle starting after a reset crosses u in the first block, or the initial cycle crosses before h and a further crossing occurs after its first reset. We bound these two possibilities separately.

A crossing in the first block that belongs to a cycle starting at a reset time $s < h$ is bounded by an unreset crossing over $[0, h-s]$. Campbell's formula for the reset epochs and independence of B and N_ν therefore bound the probability of any such crossing by

$$\nu\int_0^h \psi(u,h-s)\,ds = o(\psi(u,h)), \qquad u\to\infty. \tag{42}$$

For the last equality Lemma 5.1 implies that $\psi(u,h-s)/\psi(u,h)$ tends to zero as $u\to\infty$ for each $s\in(0,h)$. The ratio is bounded by one. In particular, separating the initial cycle from cycles starting at resets yields

$$0\le Q_\nu(u,h)-\mathbb{E}\{e^{-\nu\tau_u}\mathbf{1}_{\{\tau_u<h\}}\}\le \nu\int_0^h \psi(u,h-s)\,ds=o(\psi(u,h)).$$

By Lemma 5.1, the expectation is asymptotic to $e^{-\nu h}\psi(u,h)$, proving the one-block formula directly.

It remains to consider paths whose initial cycle crosses u before h, that is, the event $A_0=\{\tau_u<\min(h,\zeta_1)\}$. On $\{\zeta_1>T\}$, every crossing by time T precedes the first reset, so E_u implies M_u. Thus, on $(E_u\setminus M_u)\cap A_0$, necessarily $\zeta_1\le T$ and at least one required crossing occurs after ζ_1. For a process starting from zero the probability $Q_\nu(u,T)$ of any reset-process crossing by T satisfies

$$Q_\nu(u,T)\le \psi(u,T)+\nu\int_0^T\psi(u,T-s)\,ds\le(1+\nu T)\psi(u,T).$$

The event $A_0\cap\{\zeta_1\le T\}$ is measurable at ζ_1. Regeneration at this reset thus yields

$$\begin{aligned}\mathbb{P}\{(E_u\setminus M_u)\cap A_0\}&\le \mathbb{P}\{A_0\}Q_\nu(u,T)\\&\le \psi(u,h)(1+\nu T)\psi(u,T)=o(\psi(u,h)),\qquad u\to\infty.\end{aligned}$$

Together with (42), this proves the claimed negligible remainder.

Finally, writing $g_0(x;h)=\mathbb{P}\{\theta_x<h\}$, we have

$$\exp(-\nu h)g_0(x;h)\le g_\nu(x;h)\le g_0(x;h).$$

Conditioning the final unreset block on $Y(a_m)$ shows that replacing g_ν by g_0 in (15) yields $G_m(h,c)$ also for $m=2$. This proves (17) and strict positivity. Combining (41), the negligible remainder, and $\psi(u,h)\sim 2\Psi((u+ch)/\sqrt{h})$ proves (16). Formula (19) follows from $Y(h)\sim N(-ch,h)$. □

For the following lemma, let $Y_i(t)=W_i(t)-d_it$, $i=1,2$, where $d_1,d_2\in\mathbb{R}$ and $\rho\in(-1,1)$.

Lemma 5.3. *If $0<r<s$, then*

$$\mathbb{P}\left\{\sup_{t\in[r,s]}\min(Y_1(t),Y_2(t))=0\right\}=0. \tag{43}$$

Consequently, if $\eta_n=(\eta_{n,1},\eta_{n,2})\to(0,0)$ as $n\to\infty$ and $I_j=[r_j,s_j]$, $j=1,\dots,\ell$, with $s_j>r_j>0$, then

(44)

$$\mathbb{P}\left\{\bigcap_{j=1}^{\ell}\{\exists t\in I_j: Y_1(t)>\eta_{n,1},\ Y_2(t)>\eta_{n,2}\}\right\}\xrightarrow[n\to\infty]{}\mathbb{P}\left\{\bigcap_{j=1}^{\ell}\{\exists t\in I_j: Y_1(t)>0,\ Y_2(t)>0\}\right\}.$$

Moreover, for every $s>0$

$$\mathbb{P}\{\exists t\in[0,s]: Y_1(t)>\eta_{n,1},\ Y_2(t)>\eta_{n,2}\}\xrightarrow[n\to\infty]{}1. \tag{45}$$

Proof. For $I=[r,s]$, set

$$F_I(\eta):=\sup_{t\in I}\min(Y_1(t)-\eta_1,Y_2(t)-\eta_2)$$

and hence we have

$$\left|F_I(\eta)-F_I(\vartheta)\right|\le\|\eta-\vartheta\|_\infty. \tag{46}$$

To prove (43), condition on the increments of (W_1, W_2) after time r. Given these increments define

$$g_i(v) \coloneqq Y_i(r+v) - Y_i(r), \qquad 0 \le v \le s-r, \quad i = 1, 2,$$

and set

$$z = (Y_1(r), Y_2(r)).$$

Thus, under the conditioning, $g = (g_1, g_2)$ is a deterministic continuous function. Moreover,

$$F_I(0) = f_g(z),$$

where

$$f_g(z) \coloneqq \sup_{0 \le v \le s-r} \min(z_1 + g_1(v), z_2 + g_2(v)).$$

Since

$$f_g(z + t(1,1)) = f_g(z) + t, \qquad t \in \mathbb{R},$$

the zero set of f_g meets each line parallel to $(1,1)$ in at most one point and therefore has two-dimensional Lebesgue measure zero. The vector $(Y_1(r), Y_2(r))$ is independent of the post-r increments and has a non-degenerate Gaussian density. Hence, conditional on the post-r increments,

$$\mathbb{P}\{F_I(0) = 0 \mid \text{post-}r \text{ increments}\} = 0.$$

Integrating over the post-r increments proves (43). The convergence in (44) now follows from (46), the no-atom statement and dominated convergence.

It remains to consider an interval starting at zero. Put $t_n = 2^{-n}$ and

$$A_n = \{\min(W_1(t_n), W_2(t_n)) > \sqrt{t_n}\}.$$

Brownian scaling and non-degeneracy yield $\mathbb{P}\{A_n\} = p \coloneqq \mathbb{P}\{\min(W_1(1), W_2(1)) > 1\} > 0$, so $\mathbb{P}\{\limsup_n A_n\} \ge p$. This limsup belongs to the Brownian germ sigma-field; Blumenthal's zero–one law [19, Section 2.7] therefore yields it probability one. Since $d_i t_n = o(\sqrt{t_n})$, both $Y_i(t_n)$ are positive for infinitely many n, almost surely. Thus $F_{[0,s]}(0) > 0$ almost surely for every $s > 0$, and (46) yields (45). □

Proof of Theorem 3.3. Write

$$\boldsymbol{b} = (1, a)^\top, \qquad \boldsymbol{c} = (c_1, c_2)^\top, \qquad \Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad \boldsymbol{\lambda} = \Sigma^{-1}\boldsymbol{b}.$$

Both components of $\boldsymbol{\lambda}$ are positive, since $a > \rho$ and $|a\rho| < 1$. Put $\boldsymbol{\alpha}_0 = \boldsymbol{\lambda}/h$ and define

$$p_h(\boldsymbol{z}) = \mathbb{P}\{\exists t \ge 0 : W_i(t) - b_i t/h > -z_i,\ i = 1, 2\}, \qquad \boldsymbol{z} = (z_1, z_2)^\top.$$

We first show that this kernel is exactly the conditional first-block crossing probability. Given $\boldsymbol{X}(h) = u\boldsymbol{b} + \boldsymbol{z}/u$, set $L = hu^2$. For $0 \le s < L$, the reversed and scaled process $u\{\boldsymbol{X}(h - s/u^2) - u\boldsymbol{b}\}$ is Gaussian with mean $(1 - s/L)\boldsymbol{z} - \boldsymbol{b}s/h$ and covariance $\Sigma(\min(s,t) - st/L)$. Its centred part therefore has the law of

$$(1 - s/L)\boldsymbol{W}\left(\frac{s}{1 - s/L}\right), \qquad 0 \le s < L,$$

as follows directly by comparing covariances. Dividing by $1 - s/L > 0$ and using the bijection $t = s/(1 - s/L)$ yields the exact identity

$$\mathbb{P}\{\exists r \in [0, h] : X_i(r) > b_i u,\ i = 1, 2 \mid \boldsymbol{X}(h) = u\boldsymbol{b} + \boldsymbol{z}/u\} = p_h(\boldsymbol{z}). \tag{47}$$

The excluded endpoint $s = L$ corresponds to original time zero, where $X_1(0) = 0 < u$, so it cannot contribute a crossing. In particular, the conditional kernel is independent of u and of $\boldsymbol{c}$.

We next establish the integrability needed below. For $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)^\top > 0$, put

$$\Delta(\boldsymbol{\alpha}) = \frac{\boldsymbol{\alpha}^\top \boldsymbol{b}}{h} - \frac{1}{2}\boldsymbol{\alpha}^\top \Sigma \boldsymbol{\alpha}.$$

If $\Delta(\boldsymbol{\alpha}) > 0$, a union bound over unit intervals and Tonelli's theorem yield

$$I(\boldsymbol{\alpha}) := \int_{\mathbb{R}^2} p_h(\boldsymbol{z}) e^{-\boldsymbol{\alpha}^\top \boldsymbol{z}}\, dz_1\, dz_2 \le \frac{J_{\boldsymbol{\alpha}}}{\alpha_1 \alpha_2} \sum_{n=0}^{\infty} e^{-n\Delta(\boldsymbol{\alpha})} < \infty, \tag{48}$$

where

$$J_{\boldsymbol{\alpha}} = \mathbb{E}\left\{ \exp\left(\sum_{i=1}^{2} \alpha_i \sup_{0\le t\le 1} \{W_i(t) - b_i t/h\} \right) \right\} < \infty.$$

Indeed, a simultaneous crossing on $[n, n+1]$ implies that both coordinatewise maxima exceed their respective levels. These maxima are the sum of $\boldsymbol{W}(n) - n\boldsymbol{b}/h$ and an independent vector of unit-interval maxima. Their exponential moment is $J_{\boldsymbol{\alpha}} e^{-n\Delta(\boldsymbol{\alpha})}$. Integrating the two coordinatewise exceedance inequalities yields the factor $(\alpha_1\alpha_2)^{-1}$. Finiteness of $J_{\boldsymbol{\alpha}}$ follows from the reflection principle and Cauchy–Schwarz. Since

$$\Delta(\boldsymbol{\alpha}_0) = \frac{\boldsymbol{b}^\top \Sigma^{-1} \boldsymbol{b}}{2h^2} > 0,$$

the bound holds throughout a neighbourhood of $\boldsymbol{\alpha}_0$. The substitutions $\boldsymbol{z} = -h(x,y)^\top$, $t = h^2 v$, and Brownian scaling yield

$$I(\boldsymbol{\alpha}_0) = h^2 C_{a,\rho}. \tag{49}$$

This proves finiteness of $C_{a,\rho}$. The region $x < 0, y < 0$ contributes $(\lambda_1\lambda_2)^{-1}$ to its defining integral already at time zero, proving strict positivity.

Let E_u denote the simultaneous block event and let $\widetilde{Y}_i(t) = W_i(h+t) - W_i(h) - c_i t$. Conditional on the endpoint, the probability of the remaining blocks is

$$q_u(\boldsymbol{z}) = \mathbb{P}\left\{ \bigcap_{\ell=0}^{m-2} \left\{ \exists t \in [\ell h, (\ell+1)h] : \widetilde{Y}_i(t) > -z_i/u,\ i = 1, 2 \right\} \right\}.$$

By Lemma 5.3, the $\ell = 0$ event has probability tending to one, while the probability of the intersection for $\ell = 1, \dots, m-2$ tends to $\mathcal{H}_m(h, c_1, c_2, \rho)$. Dropping the $\ell = 0$ condition changes the probability by at most its failure probability, which tends to zero. Thus, also for $m = 2$ as $u \to \infty$

$$q_u(\boldsymbol{z}) \longrightarrow \mathcal{H}_m(h, c_1, c_2, \rho) \quad \text{for every fixed } \boldsymbol{z}. \tag{50}$$

The density of $\boldsymbol{X}(h)$ is

$$f_h(\boldsymbol{v}) = h^{-1} \varphi_\rho\Big((v_1 + c_1 h)/\sqrt{h}, (v_2 + c_2 h)/\sqrt{h} \Big).$$

Its ratio $\mathcal{R}_u(\boldsymbol{z}) = f_h(u\boldsymbol{b} + \boldsymbol{z}/u)/f_h(u\boldsymbol{b})$ equals

$$\mathcal{R}_u(\boldsymbol{z}) = \exp\left(-\boldsymbol{\alpha}_0^\top \boldsymbol{z} - \frac{\boldsymbol{c}^\top \Sigma^{-1} \boldsymbol{z}}{u} - \frac{\boldsymbol{z}^\top \Sigma^{-1} \boldsymbol{z}}{2hu^2} \right). \tag{51}$$

Conditional independence of the bridge before h and the increments after h, together with (47), yields

$$\mathbb{P}\{E_u\} = \frac{f_h(u\boldsymbol{b})}{u^2} \int_{\mathbb{R}^2} p_h(\boldsymbol{z}) q_u(\boldsymbol{z}) \mathcal{R}_u(\boldsymbol{z})\, dz_1\, dz_2. \tag{52}$$

Choose $\varepsilon > 0$ so small that all four vectors $\boldsymbol{\alpha}_0 - \varepsilon\boldsymbol{\sigma}$, $\boldsymbol{\sigma} \in \{-1,1\}^2$, have positive components and positive Δ. For all sufficiently large u

$$\mathcal{R}_u(\boldsymbol{z}) \le e^{-\boldsymbol{\alpha}_0^\top \boldsymbol{z} + \varepsilon(|z_1|+|z_2|)} \le \sum_{\boldsymbol{\sigma}\in\{-1,1\}^2} e^{-(\boldsymbol{\alpha}_0 - \varepsilon\boldsymbol{\sigma})^\top \boldsymbol{z}}.$$

After multiplication by p_h, the last expression is integrable by (48). Since $0 \le q_u \le 1$, dominated convergence in (52), followed by (49), yields

$$\mathbb{P}\{E_u\} \sim \frac{f_h(u\boldsymbol{b})}{u^2}\mathcal{H}_m(h, c_1, c_2, \rho)h^2 C_{a,\rho}.$$

Substituting the endpoint density proves (25). □

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